Improvement of Smartwheel Racing Wheel Data Collection & Transmission System

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INTRODUCTION

The wheelchair sports analytics sector is a fast-growing market, valued at 1.71 billion USD as of 2026, and is expected to grow to 3.59 billion USD by 2034 [1] as interest the biomechanics of wheelchair sports grows among coaches, fans, and the scientific community. There is a growing demand to understand how user-wheelchair interaction may increase the risk of common upper-limb and wrist injuries in wheelchair athletes [2]. However, biomechanic analysis is limited by a lack of available wheelchair-specific force measurement technology.



Figure 1: Professional wheelchair athletes experience substantial forces on their upper body & limbs during performance [3]

First conceived of in 1989, the Smartwheel project aims to address this scarcity by developing an instrumented racing wheelchair wheel, fitted with force, inertial, and angular position sensors to build a comprehensive picture of the mechanical interactions between an athlete and their wheelchair [4]. Since its original design, the Smartwheel has undergone decades of iterations facilitated by global advancements in force and inertial measurement technologies, digital signal processing, wireless communication, and athletic wheelchair design. Wit every iteration, more comprehensive data on biomechanical athlete-wheelchair interaction is obtained, improving the future of wheelchair athlete health, safety, and performance.

OBJECTIVES

The Summer 2026 iteration of the Smartwheel project aims to utilize modern wireless communication, serial-peripheral communication, and sensor technologies to improve data collection rate, range, and validity. Desired improvements on the Smartwheel system are as follows.

- Sampling rate was capped at approximately **70Hz** due to data output limitations of the sensors; desired rate is **1kHz**, consistent between both wheels
- Maximum connection distance was limited to **10-30m** due to Bluetooth communication limitations; desired range is **100m**, typical of the cross-distance of a racetrack
- Data needed to be **downloaded from an SD card** between runs; desired design would **stream data directly to laptop** for immediate post-processing



Figure 2: Side-by-side comparison of current work-in-progress Smartwheel (left) and previous iteration (right)

MATERIALS & METHODS

Processing & Communication

To achieve a consistent 1kHz sampling rate and minimize the latency between sensor readings and wireless communication, the dual-core architecture of the ESP32-S3 microcontroller was leveraged to use FreeRTOS parallel task-switching. Core 0 was dedicated to data collection and on-card backup data writing, driven by a 1hKz hardware timer; Core 1 was dedicated to WiFi communication and command handling, driven by incoming commands from the laptop. Tasks were delegated to prioritize incoming commands, then rapid data collection/transmission, and lastly backup data writing. Data is continuously added to a shared dataQueue structure, which is emptied by the transmitter task upon receiving an appropriate command. This ensures seamless access to data when a data collection run is active, while minimizing blocking tasks that would interfere with data collection.

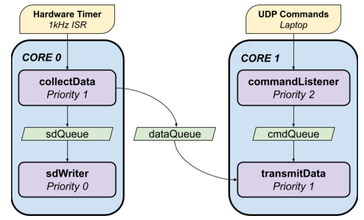


Figure 3: General layout of task delegation & data structure sharing between cores

Communication between the ESP32 and user's laptop is a UDP protocol; this ensures rapid data packet transmission without the latency of communication handshakes requires by other protocols such as TCP, while allowing for greater connection range compared to Bluetooth.

Hardware & Peripheral Selection

The main microcontroller unit of the previous Smartwheel design, a Teensy 4.1, was replaced with an ESP32-ProS3 module for dual-core capabilities and integrated WiFi communication. All peripherals, excluding the microSD module, were replaced to match the objectives of the project.

Load Cell ADC	IMU	Encoder
ADS131M08	LSM6DSO32	AMT222B-V
8 simultaneously sampling differential inputs Programmable data rate up to 32 kSPS Programmable gain up to 128 Low-drift internal voltage reference	±4/±8/±16/±32 g full scale ±125/±250/±500/±1000/±2000 dps full scale SPI / I²C & MIPI I3CSM serial interface with main processor data synchronization	12 or 14-bit absolute position with multi-turn capability checksum bits for error detection digitally settable zero position

Table 1: Overview of hardware selection to match 1kHz low-noise specifications for Smartwheel redesign

Usability

In addition to improving the data transmission & communication system, great improvements in the usability of the Smartwheel system were also made. A full MATLAB-based graphic user interface was implemented, which guides the user through selecting active wheels, connecting via WiFi, synchronizing the timestamps of the data packets to the laptop's absolute UTC time, automatically synchronizing runs to a VICON motion capture system, and calibrating the initial angular offset of the wheels. This more usable interface is intended to reduce researcher's stress and frustration and minimizes the chance of human error impacting data validity.

RESULTS

Data Accuracy

Rough qualitative validation of Smartwheel data was performed by collecting trial run data and comparing the correlation between each wheel's z-axis gyroscope readings (indicating rad/sec about the z-axis) and the rate of change of encoder positions. *Table 2* and *Figure 3* show significant discrepancy between the left and right wheels; while the right Smartwheel data shows consistency with encoder readings at >0.98 correlation, left Smartwheel IMU data shows a lot more noise at <0.53 correlation. This agrees with the percentage of gyroscope reading outliers detected in the Smartwheel data after applying a Hampel-style spike filter. Consistently, the left Smartwheel gyroscope shows significant noise (~11.52% in all axes) compared to the nearly ideal noise (<1%) of the right wheel.

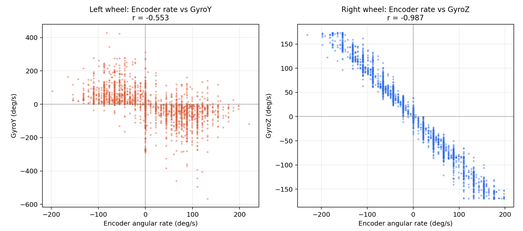


Figure 3: Visualization of correlation between Z-axis gyroscope data and encoder readings from left and right Smartwheel. AI Use Disclaimer: This figure was generated with the help of Claude AI using real validated data.

Wheel	Outlier % (GyroZ)	Outlier % (AccelZ)	Outlier % (Mean, all 3 gyro axes)	r (GyroZ vs. encoder rate)
Right	0.62%	1.40%	0.87%	−0.987
Left	8.01%	16.86%	11.52%	0.526

Table 2: Comparison of Smartwheel gyroscope-encoder correlation to overall gyroscope outlier detection

Data Quality

To assess the quality of data transmission/collection (not necessarily accuracy), the post-processing data cleaning script written to process raw Smartwheel data produces a report detailing the number of dropped data packets (relative to the ideal 1kHz transmission) as well as outliers.

Packet Loss	Left	Right
Unique samples captured	16,564	19,335
Raw packets received (incl. duplicates	20,170	21,490
Missing/generated (interpolated)	4,943	2,172
Loss rate (% of 1kHz grid)	22.98%	10.10%

Table 3: Comparison of Smartwheel packet loss & generation between left and right wheels

These results are consistent with the previous reports on gyroscope noise; the right Smartwheel shows acceptable data packet loss for a prototype stage (~10%) whereas the left Smartwheel shows considerable loss (~22%).

Furthermore, the quality of the load cell data provided by the Smartwheel system was not formally quantified, but it was observed that several channels (namely CH3A-L and CH3/4-R) experiences consistent hardware wiring issues. These wiring issues will be addressed, but it should be stated that load cell data could not yet be validated.

CONCLUSIONS

The updated design of the Smartwheel system represents a very early-stage prototype of a very promising addition to the legacy of the project. From certain trial runs collected from right-wheel data, low-noise, high-data-rate, low-loss positional and force data can be obtained from improved sensors and parallel-process firmware. However, there are several issues that must be addressed in future iterations of this design:

SPI Bus Contention: During the hardware design phase of this project, it was not known that SPI contention would be an issue across the peripherals. Although tasks are split across two cores, all four SPI peripherals contend for one SPI bus. This is a simple redesign in future iterations – it is recommended to move the SD module to its own hardware bus, as the provided data in the *RESULTS* section was only possible by eliminating SD card writes.

Left/Right Wheel Discrepancy: Clear differences in data quality and accuracy can be observed. This illuminates the challenge of creating two identical wheels which each may experience different hardware/wiring issues. However, the promising results from the right Smartwheel also shows the possibility of negligible data loss/corruption at high wireless data rates.

Load Cell Calibration: The crucial aspect of the Smartwheel system which was not focused on in the 2026 iteration is the interpretation of load cell data. As-is, the Smartwheel system streams raw load cell readings to the user's laptop, to allow for future post-processing and interpretation of that data. This will require a formal calibration sequence and force vector resolution post-processing script, which is still in development.

Usability: A highlight of this project's outcomes, the usability and quality of documentation for the Smartwheel project has been much improved. The entire data collection procedure has been streamlined into one WiFi setup → connection → synchronization → z-offset calibration → VICON sync → collect data pipeline, all achieved through a graphic MATLAB interface. While data accuracy and validation remains the most crucial aspect of this project, having the working prototype of the Smartwheel system be easier to navigate and learn reduces the risk of human error and allows for easier future iteration of the project.

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